

SMARTBRIDGE

Let's Bridge the Gap | SKILL WALLET - A SmartBridge Product

Date	29-09-2026
Team ID	SWTID-2026-7949
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value

Code-Layout, Readability and Reusability

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes/No/Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Standard 2/4-space indentation adhered to across all ACL script blocks and IIFE structures.
2	Proper File Structure	Files and folders are logically organized	Yes	ACL rules cleanly grouped in 'ACL Project [Global]' update set and modular ServiceNow sys_security_acl records.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Self-descriptive naming adopted for custom roles (bb1, bb2, bb3, bb4) and fields (u_institution_details).
4	Function/Method Names	Functions are descriptively named	Yes	Standard ServiceNow Glide APIs utilized cleanly: gs.hasRole('admin') and gs.hasRole('bb1').
5	Code Comments	Inline and block comments explain logic	Yes	Explicit inline comments explain admin full-access override, branch evaluation, and fallback access denial.
6	Modular Design	Code is split into reusable functions/modules	Yes	Independent, modular ACLs constructed for CRUD operations (Read, Create, Write, Delete).
7	No Redundant Code	Duplicate or unused code is removed	Yes	Streamlined JavaScript IIFE execution; minimal overhead without redundant queries or unused variables.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Fail-safe default 'return false' ensures zero-trust denial when role/condition

			criteria fail.
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Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High/Med/Low)
1	Script-Controlled ACL Evaluation Script (IIFE)	Server-Side JavaScript (ES512 mode)	u_institution_details Read ACL & future branch tables	High
2	Role Authorization Check Helper (gs.hasRole)	ServiceNow GlideSystem API	Read, Create, Write, and Delete ACL condition pipelines	High
3	Custom Security Role Definitions (bb1 - bb4)	ServiceNow RBAC / sys_user_role	Departmental user groups, ACL requires_role list	High
4	Custom Schema Definition (u_institution_details)	ServiceNow System Definition / Tables	Student details, institutional department lists	Medium
5	Global Update Set Container (ACL Project)	ServiceNow Update Set Framework	Dev-to-Prod deployment package & XML artifact export	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5 / 5	Strictly structured using ServiceNow scoped script standards, self-contained IIFE execution, and proper condition separation.
Readability	5 / 5	Code is clean, concise, intuitive, and includes explicit conditional branching for administrative and departmental role paths.
Reusability	4.8 / 5	Modular ACL patterns and role schemas can be replicated seamlessly across any other academic branches (ECE, CSE, MECH).
Documentation / Comments	4.9 / 5	Comprehensive inline comments detail privilege verification, field match logic, and fail-safe denial.
Overall Score	4.9 / 5	Enterprise-ready implementation complying with ServiceNow security administration best practices and row-level access standards.